

# Identifying new earnings management components: a machine learning approach

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## Abstract

**Purpose** – This study aims to apply machine learning (ML) to identify new financial elements managers might use for earnings management (EM), assessing their impact on the Standard Jones Model and modified Jones model for EM detection and examining managerial motives for using these components.

**Design/methodology/approach** – Using eXtreme gradient boosting on 23,310 the US firm-year observations from 2012 to 2021, the study pinpoints nine financial variables potentially used for earnings manipulation, not covered by traditional accruals models.

**Findings** – Cost of goods sold and earnings before interest, taxes, depreciation and amortization are identified as the most significant for EM, with relative importances of 40.2% and 11.5%, respectively.

**Research limitations/implications** – The study's scope, limited to a specific data set and timeframe, and the exclusion of some financial variables may impact the findings' broader applicability.

**Practical implications** – The results are crucial for researchers, practitioners, regulators and investors, offering strategies for detecting and addressing EM.

**Social implications** – Insights from the study advocate for greater financial transparency and integrity in businesses.

**Originality/value** – By incorporating ML in EM detection and spotlighting overlooked financial variables, the research brings fresh perspectives and opens new avenues for further exploration in the field.

**Keywords** Earnings management, Machine learning, Accrual manipulation, Gradient boosting algorithm

**Paper type** Research paper



## 1. Introduction

Since the late 1980s, scholars extensively studied earnings management (EM), developing methods like the Standard Jones Model (SJM) by Healy (1985) and Jones (1991). SJM assumes identifiable nondiscretionary accruals, using deviations to spot discretionary accruals indicating EM. Dechow and Dichev (2002) refined SJM into the modified Jones model (MJM). Yet, accrual-based models face criticism for their limited predictive ability (Almasarwah, 2019; Jackson, 2018) and complexity, hindering comprehension (Connolly and Hyndman, 2006).

Moreover, reliance on financial data in these models risks manipulation or errors (Beneish, 1997).

EM manipulates financial statements to meet targets through upward or downward techniques. Upward EM inflates earnings to meet forecasts or analyst expectations, while downward EM reduces earnings to manage expectations or avoid scrutiny. Detecting EM is crucial for investor confidence and financial reporting reliability, leading to stricter global regulations. However, existing detection methods are limited, emphasizing the need for advanced techniques through further research and development (R&D).

Recent advancements in accounting and finance have introduced machine learning (ML) techniques for detecting and analyzing EM practices effectively. ML algorithms compare deviations from past trends or peer data to uncover irregularities indicating earnings manipulation. Trained on historical financial data, these algorithms recognize EM characteristics and apply them to new data for detection, as shown by research (Hammami and Hendijani Zadeh, 2022). Integrating ML into EM evaluation offers a more efficient way to identify manipulation compared to traditional methods (Dbouk and Zaarour, 2017).

The primary objective of this research is to use ML to explore new financial factors that managers might exploit for earnings manipulation via accruals. We aim to answer:

- Q1. How do additional financial elements affect the detection ability of SJM and MJM in identifying EM?
- Q2. What drives managers to manipulate earnings using these elements?

This study analyzed a data set comprising 23,310 observations of the US-listed firms from 2012 to 2021. Our aim was to uncover new components of EM by training ML models to explore the relationship between nine variables and the outputs of SJM and MJM. We used eXtreme gradient boosting (XGBoost), a robust ML algorithm known for enhancing predictive accuracy. Using XGBoost as a classification model, we predicted SJM and MJM outcomes based on our chosen variables.

Results reveal that the cost of goods sold (COGS) holds the greatest significance in EM projections, with an average importance of 40.2%. Managers manipulate COGS to adjust earnings both upward and downward to meet targets. Earnings before interest, taxes, depreciation and amortization (EBITDA) follow as the second most crucial factor, primarily influenced by downward EM. Additionally, Total Debt (TD), Total Shareholders' Equity (TE) and R&D expenses significantly impact EM prediction. While less influential, Retained Earnings (RE) and Net Income (NI) also contribute to improving accruals-based models' ability to detect EM across most data sets. Incorporating RE and NI into these models could thus play a vital role in EM detection.

This study contributes to the existing literature in two ways. First, this study contributes to the literature of detecting EM through an approach that integrates predictive analytics with audit procedures. This study is motivated by the various advantages of leveraging ML models within the audit field. Using ML algorithms as earnings manipulation detection tools can help forensic accountants, auditors and regulators uncover hidden opportunistic practices. Compared to conventional linear regression models, adopting ML methodologies can enhance the accuracy of detection.

The proposed method tackles EM detection challenges, aiding early identification of at-risk firms, benefiting stakeholders. Integrating technology with auditing keeps the industry ahead of risks. This study's results are crucial for audit professionals and policymakers, underlining the need for innovative methods to enhance financial report quality and transparency. It also boosts stakeholder awareness and defense against EM strategies.

Additionally, the research advances anti-EM efforts by uncovering new factors and their effects, leading to better detection models. Emphasizing ML algorithms in audits marks a move toward advanced practices, enhancing fraud detection capabilities and underscoring a dedication to ethical financial standards, thereby increasing transparency and safeguarding financial report integrity.

Second, our research augments the effectiveness of accruals-based models in detecting earnings manipulation by integrating supplementary financial data pertaining to factual economic activities. Although accruals-based models are extensively used to identify potential earnings manipulation practices, they are limited in scope as they concentrate solely on accounting entries, making it difficult to uncover earnings manipulation involving authentic economic transactions.

To overcome this limitation, our findings suggest the integration of pertinent financial data concerning actual transactions to augment the precision of accruals-based models in detecting earnings manipulation. By doing so, these models can identify even minor alterations in a company's financial performance that may indicate earnings manipulation. By incorporating a new component into accruals-based models, a more comprehensive approach to detecting and understanding potential earnings manipulation practices could be achieved.

Finally, while acknowledging the importance of nonfinancial aspects of EM highlighted in existing literature (Almahrog *et al.*, 2018; Prior *et al.*, 2008; Riahi-Belkaoui, 2003), this study exclusively examines its financial dimensions. Our objective is to construct a robust analytical framework centered on how financial factors influence companies' earnings reporting practices. By concentrating solely on these financial determinants, we seek to identify EM patterns with greater precision, yielding more dependable findings. This focused approach enables us to offer practical recommendations for financial decision-making and reporting enhancement. Furthermore, it facilitates a deeper exploration of the root causes of EM and its repercussions on corporate governance and investor confidence. While we acknowledge the broader scope of nonfinancial aspects, our study deliberately leaves their exploration to future research endeavors.

The rest of the paper is organized as follows: Section 2 provides a brief theoretical background on the commonly used EM theories. In Section 3, we discuss our variable selections and formulate our hypotheses. Section 4 describes our research design, while Section 5 provides an overview of our data and the performance of our ML model (XGBoost), along with the presentation of our empirical findings. Finally, in Section 6, we conclude our study.

## 2. Theoretical background

Investigating EM is crucial for ensuring financial reporting accuracy, protecting investors, meeting regulatory standards, enhancing transparency, accountability and corporate governance, thus avoiding damage to a company's reputation and stakeholder trust. EM involves manipulating company income through discretionary accruals, with *agency* and *institutional* theories offering explanations (Kury, 2007). Agency theory highlights the conflict between shareholders (principals) and managers (agents), suggesting managers might manipulate earnings to serve their interests, potentially harming owners (Harrison and Harrell, 1993; Harrell and Harrison, 1994; Tuttle *et al.*, 1997; Rutledge and Karim, 1999; Booth and Schulz, 2004). It proposes using monitoring and incentives, like performance-based pay, to align managers' actions with shareholders' goals, promoting ethical reporting.

In contrast, institutional theory points to the role of external pressures, such as legal standards and societal expectations, in shaping EM practices. It suggests companies might engage in EM to align with these external benchmarks, aiming for legitimacy and social acceptance (Greenwood *et al.*, 2011; Pache and Santos, 2010; Powell and DiMaggio, 2012). However, it also notes the potential conflict between adhering to external expectations and internal values, such as social responsibility, illustrating the complex interplay between organizational goals and societal standards in financial reporting (Zhao *et al.*, 2017).

### 3. Literature review and hypotheses development

To identify EM, an ML model can be developed using financial statements labeled with key information. Once trained, this model can efficiently predict EM in new financial statements, aiding analysts and auditors in quicker detection. This method uses various ML algorithms like decision trees, random forests and support vector machines, which have been proven effective in past research (Gerakos, 2012; Kang and Park, 2021).

Accrual models' examination has revealed their strengths and weaknesses. As highlighted by Hammami and Hendijani Zadeh (2022), assessing accrual models' accuracy by comparing them with actual financial statement figures is crucial for their improvement. This comparison not only benchmarks accuracy but also guides model refinement. On the other hand, Chen *et al.* (2022) introduced a novel approach using ML to predict future earnings from comprehensive financial data, diverging from traditional forecasting methods. Their analysis showed ML's superiority in capturing financial nuances and forecasting accuracy, underscoring its potential to enhance financial trend predictions.

ML is increasingly applied in fraud detection, with Rahul *et al.* (2018) introducing a method to spot earnings manipulation through advanced algorithms, enhancing the ability to identify fraud in financial reports and thus bolstering market integrity. Similarly, Sanad (2021) applies ML to detect EM, contributing to greater financial transparency and underscoring the role of technology in modern financial analysis.

Yet, accrual models face criticism for their limited predictive power and complexity, potentially baffling those without finance or accounting knowledge (Almasarwah, 2019; Jackson, 2018; Chen *et al.*, 2022). Their dependency on financial data, which can be manipulated, also questions their reliability (Beneish, 1997).

Research identifies improvements needed in accrual models, highlighting their occasional distortion of economic realities, sensitivity to accounting choices and omission of cash flow data. These models also suffer from limited generalizability due to assumptions not reflecting real-world conditions and vulnerability to changes in variable selection or model specification, questioning their robustness (McNichols and Stubben, 2018; Ogbaisi and Aronmwan, 2022; Dbouk and Zaarour, 2017; Purba *et al.*, 2022; Dechow and Dichev, 2002; Dechow *et al.*, 2012; Peasnell *et al.*, 2000). Despite these critiques, integrating ML presents opportunities to improve financial models' accuracy and relevance, emphasizing the need for ongoing innovation and critical evaluation in financial analytics.

Furthermore, the role of debt in a company's capital structure affects accruals models' effectiveness, with a strong correlation between financial leverage and accruals management. Firms with significant debt might manipulate earnings to meet debt covenants or appear financially sound, with this inclination more evident in highly indebted companies. This leads to a noticeable relationship between accruals and future cash flows as companies adjust earnings to fulfill debt requirements (Lee *et al.*, 2021).

To account for the impact of debt on financial reporting, accruals models now include financial leverage indicators or debt ratios, enhancing their reflection of a firm's debt on

accruals practices (Mamatzakis *et al.*, 2023). Yet, the impact of debt varies, highlighting the importance of a firm's specific context – like its industry, market conditions and financial health – in influencing accruals management.

Additionally, considering R&D expenses is crucial for refining accruals models. Models incorporating R&D expenses through measures of intensity or capitalization provide insights into how innovation investments affect accruals management (Shust, 2015). This inclusion recognizes R&D's strategic value and its effect on accruals, offering a more detailed analysis tool for analysts and investors.

The effectiveness and accuracy of accruals models are significantly influenced by the time period analyzed, as changes in firms' operations, economic conditions and regulations can affect model relevance over time. The choice of an appropriate analysis timeframe is crucial to maintain model efficacy, ensuring it captures essential information without being overwhelmed by irrelevant data (Collins *et al.*, 2017; Peasnell *et al.*, 2000).

Creating advanced accruals models demands a comprehensive approach that includes debt, R&D spending and the time dynamics of financial reporting. These models are designed not just to detect financial adjustments but to provide insights into firms' strategic financial management, helping stakeholders navigate the complex economic environment.

Finally, EM splits into upward and downward manipulation, each with unique motives and impacts. Upward EM inflates earnings, boosting stock prices and executive rewards, presenting an illusion of financial health but risking future expectations, investor trust and legal or reputational damage upon discovery (Dechow and Dichev, 2002; Dechow *et al.*, 2012). Downward EM, on the other hand, underreports earnings to avoid debt breaches or save profits for later, possibly indicating financial issues and raising borrowing costs by hiding real performance (Almasarwah, 2019). Both types of EM misrepresent financial realities, damaging financial reporting integrity and affecting stakeholder decisions. This misrepresentation highlights the need for effective detection to ensure genuine financial disclosures.

Building on the discussions of EM's implications and its detection, we propose two hypotheses aimed at enhancing the predictive power of accruals-based models through the integration of novel financial variables. These hypotheses are posited as follows:

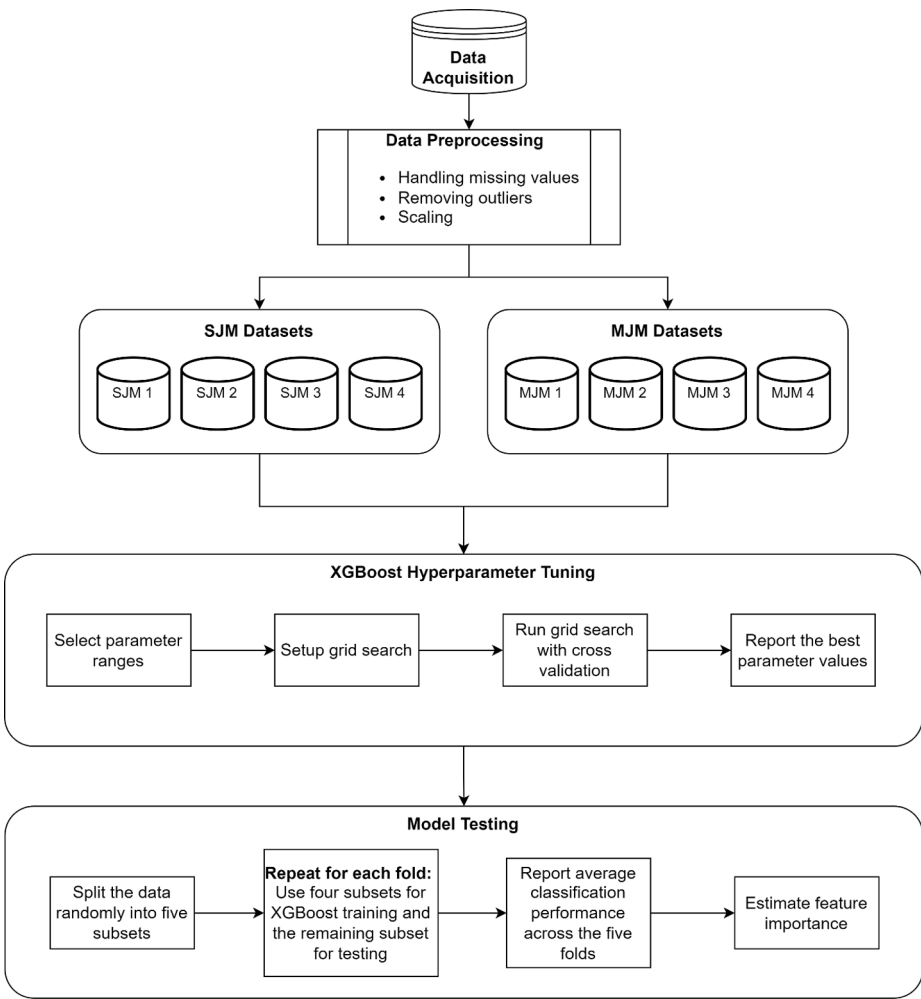
- H1. The incorporation of new financial variables into accruals-based models notably enhances their capability to identify EM activities.
- H2. The employment of additional financial variables, beyond those traditionally used in established models like the Jones model and its modifications, yields substantial evidence of both upward and downward EM by managers.

## 4. Research design and variable selection

### 4.1 Research design

The methodology used in this study for identifying novel components of EM is depicted in Figure 1. We developed a classification model, followed by an assessment of the importance of input variables using the model. The process commenced with meticulous data acquisition and preprocessing to ensure its suitability for classification purposes. We estimated the relative importance of each input variable based on its parameters in the model.

The first step of our methodology is to derive a response variable from SJM and MJM outputs [1]. These outputs were categorized into three groups representing distinct directions of EM change: 0 for low EM, 1 for high positive EM (upward EM) and 2 for high

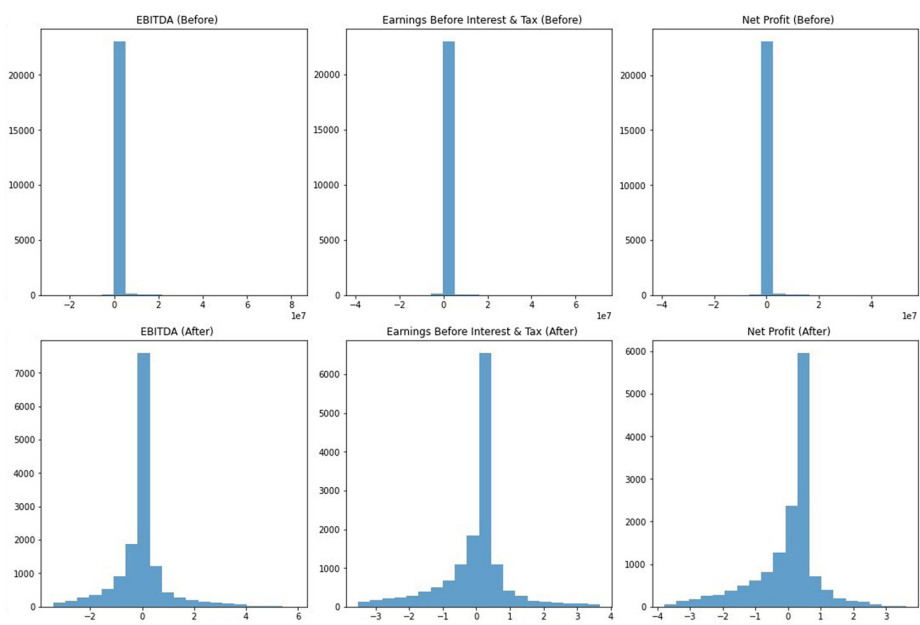


**Figure 1.**  
Research  
methodology

**Source:** Authors' own creation

negative EM (downward EM). From these groupings, we constructed four distinct data sets based on various combinations thereof, namely, (0, 1, 2), (0, 1), (0, 2) and (1, 2), yielding a total of eight data sets for both SJM and MJM.

The subsequent phase involved data preprocessing. Initially, we addressed missing data by removing corresponding samples. Following this, we performed outlier detection based on the interquartile range (IQR) of each variable, removing samples that fell outside the range of 1.5IQR below the first quartile or 1.5IQR above the third quartile. Finally, we standardized the variables using standard normal scaling. The outcomes of these preprocessing steps are visually represented in Figure 2, which illustrates the effect of outlier removal and scaling through histograms for three variables, demonstrating consistent transformations across all nine variables.



**Figure 2.**  
Histograms of  
EBITDA, earnings  
before interest and  
tax and net profit  
before and after data  
preprocessing

**Source:** Authors' own creation

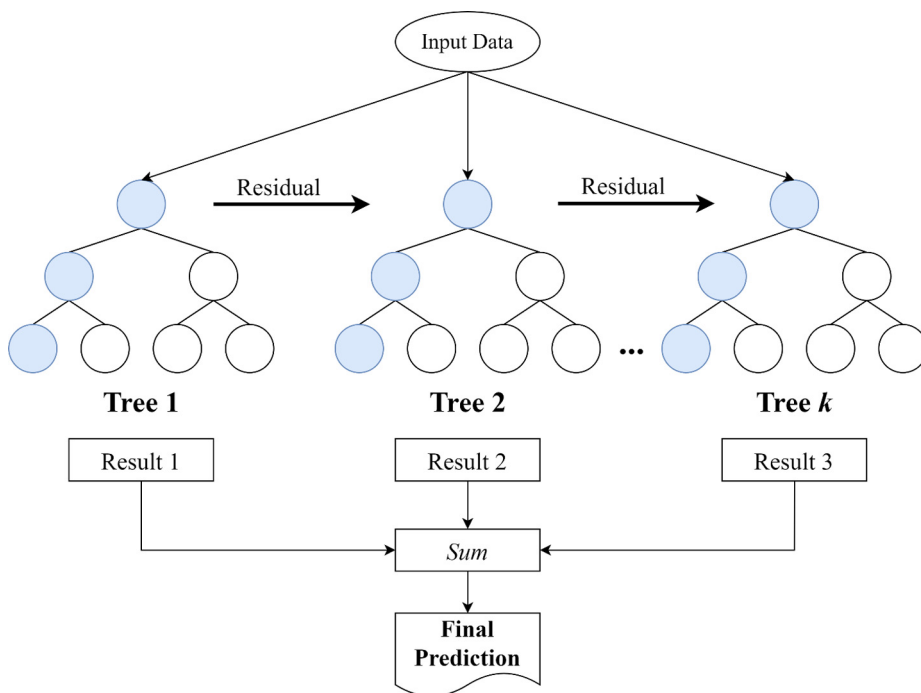
The third step involved optimizing the hyperparameters of the XGBoost model. To achieve this, we used grid search in conjunction with cross-validation to identify hyperparameter values that maximize model accuracy for each data set. This process systematically explores a grid of hyperparameter values to pinpoint the optimal combination for the model. The hyperparameters considered, along with their respective ranges, encompassed the learning rate (0.001, 0.005, 0.01, 0.05, and 0.1), maximum depth (3, 4, 5, 6 and 7) and a number of estimators (100, 150, 200, 250 and 300).

In the final step, we leveraged the best set of hyperparameters identified in the previous stage to train and test a series of eight classification models. Using XGBoost as the classification model, we predicted various levels of SJM and MJM outputs based on a new set of selected variables.

XGBoost stands out as a formidable ML algorithm renowned for its exceptional efficiency and accuracy across regression and classification tasks. Initially introduced by [Chen and Guestrin \(2016\)](#), XGBoost operates by constructing a sequence of decision trees, as shown in [Figure 3](#), that iteratively refine upon the errors (residuals) of their predecessors. Notably, the algorithm incorporates a regularization term to mitigate the risk of overfitting, ensuring the model's capacity to generalize beyond the training data set ([Montesinos López et al., 2022](#)). With its scalability tailored to handle large and high-dimensional data sets, XGBoost emerges as a versatile tool applicable across a myriad of domains, including finance, accounting and marketing ([Wang and Yang, 2021](#); [Tan et al., 2023](#); [Ben Jabeur et al., 2023](#)).

While ensemble models and decision trees have found utility in EM detection, XGBoost remains conspicuously absent from existing literature within this domain. Although a comprehensive review of ML applications in EM detection exists ([Almaqtari et al., 2021](#)), the integration of XGBoost represents a novel approach deserving of exploration.





**Figure 3.**  
XGBoost model  
structure

**Source:** Authors' own creation

At the heart of XGBoost lies the gradient boosting algorithm, encapsulated by the fundamental equation:

$$F(x) = f(x) + \sum_{t=1}^T h(x, \theta_t) \quad (1)$$

where:

- $F(x)$  is the final prediction of the model;
- $f(x)$  is the initial prediction, usually set to the average of the target variable;
- $h(x, \theta_t)$  is the  $t^{th}$  boosting tree;
- $\theta_t$  is the parameter of the  $t^{th}$  boosting tree; and
- $T$  is the total number of boosting trees.

In each iteration, a new boosting tree is incorporated into the model to rectify the errors of its predecessors. This optimization process hinges on gradient descent, an iterative optimization algorithm leveraging the gradient of the loss function to refine the model's parameters.

To gauge the classification performance of the trained XGBoost models, we use  $k$ -fold cross-validation, a robust technique ensuring reliable estimations of model efficacy [2]. Subsequently, the XGBoost models generated serve as invaluable tools for elucidating the significance of the selected variables, facilitating the identification of their relative importance in forecasting both SJM and MJM outputs.



4.2 Variables selection

Our study fills a crucial gap in EM and accruals model research by introducing variables previously not extensively considered, marking our approach as innovative. We select variables neglected or minimally used in standard models, chosen for their theoretical importance in explaining EM. Unlike traditional models that stick to well-known variables, our expanded framework includes factors suited to the changing landscape of financial reporting, aiming for a richer understanding of EM and enriching accounting and finance research.

From 19 possible variables, we selected nine that firms might use for earnings manipulation, not typically found in standard accruals models, underlining their significance to our analysis. [Table 1](#) lists these variables. We provide brief rationales for their inclusion, highlighting their relevance to EM analysis.

*Time variable (Years):* EM can be influenced by time variables such as reporting periods, accounting standards and reporting period length. During certain periods, firms may manipulate accounting numbers or use aggressive techniques to boost earnings to meet targets or manage expectations. Changes in accounting standards or regulations can also impact EM practices, while longer reporting periods can increase transparency and scrutiny.

*Total debt (TD):* The manipulation of interest expenses and off-balance sheet financing, such as operating leases or special purpose entities, can be a way in which total debt can affect EM. A company may manipulate interest payments and recognition of interest expense to increase or decrease net income. Off-balance sheet financing can also impact the debt-to-equity ratio, which affects the perception of the company's financial health.

*Total equity (TE):* Shareholders' equity can impact EM through the manipulation of stock-based compensation expenses and the amount of shareholder's equity reported on the balance sheet can be manipulated through various methods. Reclassifying expenses or income, manipulating the value of equity investments and understating or overstating the value of the company's own stock are examples of such methods.

| #  | Considered variables                                           | Final variables after running XGBoost |
|----|----------------------------------------------------------------|---------------------------------------|
| 1  | PPE                                                            | NO                                    |
| 2  | Depreciation                                                   | NO                                    |
| 3  | Total current assets                                           | NO                                    |
| 4  | Accounts receivable                                            | NO                                    |
| 5  | Fixed assets                                                   | NO                                    |
| 6  | Total assets                                                   | NO                                    |
| 7  | Total current liabilities                                      | NO                                    |
| 8  | Total liabilities and debt                                     | YES                                   |
| 9  | Total shareholders' equity                                     | YES                                   |
| 10 | Retained earnings                                              | YES                                   |
| 11 | Total revenues                                                 | NO                                    |
| 12 | Gross sales                                                    | NO                                    |
| 13 | Net sales                                                      | NO                                    |
| 14 | Cost of goods sold                                             | YES                                   |
| 15 | Research and development expenses                              | YES                                   |
| 16 | Earnings before interest, taxes, depreciation and amortization | YES                                   |
| 17 | Earnings before interest and tax                               | YES                                   |
| 18 | Net income                                                     | YES                                   |
| 19 | Time variable                                                  | YES                                   |

**Table 1.**  
Variable selection

**Source:** Authors' own creation

*Retained earnings (RE)*: retained earnings can impact EM by manipulating the amount reported on the balance sheet or by manipulating dividends paid to shareholders. Manipulating the amount of retained earnings can be achieved through activities such as improperly recording revenue or expenses, overstating or understating reserves or misrepresenting the value of assets or liabilities. Firms may also manipulate dividends paid to artificially increase or decrease NI in a given period by delaying or accelerating dividend payments.

*Cost of goods sold (COGS)*: EM can be influenced by COGS through the manipulation of inventory value and the recognition of expenses associated with COGS. By manipulating inventory value, a company can increase or decrease COGS, which affects NI and changes the perception of the company's financial performance.

*Research and development (R&D)*: R&D expenses can impact EM through the capitalization of costs and manipulation of timing and amounts of R&D expenses. Capitalizing R&D costs allows firms to defer expenses and reduce current period expenses, increasing net income. Delaying or accelerating R&D projects can also impact recognition of R&D expenses in a given period. Manipulating the amount of R&D expenses reported on the income statement can be done by misclassifying expenses or inflating expenses to manipulate net income.

*Earnings before interest, taxes, depreciation and amortization (EBITA) and earnings before interest and taxes (EBIT)*: EBITDA and EBIT are both financial metrics that can be subject to EM practices, leading to a distortion of a company's financial results and misrepresentation of its true financial health. EBITDA is commonly manipulated by excluding certain operating expenses to show stronger operating performance or meet debt covenant requirements. Meanwhile, EBIT can be manipulated by delaying the recognition of expenses or accelerating revenue recognition to give the appearance of better performance. Firms may also manipulate their EBIT results to avoid defaulting on loans or facing penalties.

*Net income (NI)*: NI can be manipulated in several ways to impact EM. Accounting practices are used to artificially inflate or decrease net income, manipulate NI to meet or exceed analyst expectations or adjust charges or expenses to hit internal targets. Furthermore, executives may have incentives based on NI performance, leading them to manipulate the company's financial results to achieve specific targets.

## 5. Results

### 5.1 Data

Our analysis began with the compilation of accounting data for all the US-based listed firms from 2012 to 2021. This timeframe was meticulously chosen, as the market only began to exhibit strong signs of recovery from the 2008 financial crisis starting in 2012. Moreover, we concluded our sample in 2021 to mitigate any potential biases that the COVID-19 pandemic could introduce. Consequently, the selection of financial data spanning from 2012 to 2021, based on US global industry classification standard codes, provides researchers with an extensive data set. Initially, we had a sample of 34,410 firm-year observations, which was then reduced by eliminating financial institutions and firms with missing data.

Initially, we had a sample of 34,410 firm-year observations, which was then reduced by eliminating financial institutions and firms with missing data. Ultimately, our final sample consisted of 23,310 firm-year observations obtained from 2,331 firms.

We based our analysis on two main data sets. The two data sets include the same set of independent variables but different values for the response variable Y. The response variable Y in the first data set was generated by SJM and by MJM in the second data set.

As mentioned earlier, a total of eight data sets out of the main two data sets were generated by slicing the data using different combinations of the response variable Y values. Table 2 provides a description of the eight data sets, including data set size and response variable values, i.e. class labels. The data sets that include SJM in their names are the ones that were generated based on SJM's Y values. Similarly, the data sets that include MJM in their names are the ones that were generated based on MJM's Y values.

5.1.1 *Performance measurements and testing setup.* In this section, we explain the performance measures used to evaluate the classification performance of the XGBoost models. We used *k*-fold cross-validation to obtain the average performance of the models that we developed based on the set of variables selected at the variable selection stage. In *k*-fold cross-validation, the data is divided into *k* subsets or “folds,” of roughly equal size. The model is trained using *k* – 1 folds and tested using the remaining fold. This process is repeated *k* times by using each fold exactly once as the testing data. The performance measures are then averaged across the *k* tests to give an overall performance metric for the model. We used five-fold cross-validation in this study. Cross-validation helps in minimizing the risk of model overfitting by testing the model on multiple independent subsets of the data.

The results of the *k* folds are averaged to give an overall performance metric for the model. After using the trained XGBoost models to predict Y values for testing data sets, we obtain counts of true positive (TP), true negative (TN), false positive (FP) and false negative (FN) predictions. We used the following measures:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{4}$$

$$F1 = 2* \frac{precision * sensitivity}{precision + sensitivity} \tag{5}$$

**Table 2.**  
Description of the  
data sets used for  
model development  
and analysis

| Data set | No. of variables | No. of samples | Response variable (Y)* (class labels) |
|----------|------------------|----------------|---------------------------------------|
| SJM 1    | 9                | 23,310         | 0, 1, 2                               |
| SJM 2    | 9                | 9,121          | 0, 1                                  |
| SJM 3    | 9                | 20,038         | 0, 2                                  |
| SJM 4    | 9                | 17,461         | 1, 2                                  |
| MJM 1    | 9                | 23,310         | 0, 1, 2                               |
| MJM 2    | 9                | 7,199          | 0, 1                                  |
| MJM 3    | 9                | 20,427         | 0, 2                                  |
| MJM 4    | 9                | 18,994         | 1, 2                                  |

**Notes:** \*Response value of 0 represents low EM, while 1 represents high positive earnings management (upward EM) and 2 represents high negative EM (downward EM)  
**Source:** Authors' own creation

These measures describe the effectiveness of classification models from different aspects. Accuracy measures the ability of a model to make correct classifications. Sensitivity measures the ability of a model to correctly predict the positive class (e.g.  $Y = 0$  can be considered positive and  $Y = 1$  can be considered negative or vice versa). Precision is the ratio of correctly classified positive samples to all samples that were classified as positive. *F1* score combines precision and sensitivity. These measures are designed for binary classification that includes two classes (e.g. SJM 2 has  $Y = 1, 2$ ). For multiclass models that include more than two classes in the response variable (e.g. SJM 1), we used macro averaging of the metrics which is a simple arithmetic average of the four metrics described earlier across the different data classes.

To compare the importance of different variables selected at the variable selection stage, we used the *Gain* importance metric available in the XGBoost Python package (XGBoost Documentation, 2023). Gain measures the relative contribution of a certain variable to an XGBoost model. The contribution here is defined as the number of time that variable was used in each tree of the model.

*5.1.2 Extreme gradient boosting model performance.* Table 3 includes classification performance of the XGBoost models that were developed for each data set. The table provides average (AVG) and standard deviation (STDEV) across the five cross-validation folds.

Results presented in Table 3 exhibit remarkable consistency across all eight data sets. The average performance measure values exceed 66%, indicating that the selected variables in each data set produced desirable prediction performance and exhibited strong associations with the SJM and MJM outputs. Moreover, we observe that the SJM 4 and MJM 4 data sets, which feature response values 1 (high positive EM) and 2 (high negative EM), achieved the highest classification performance.

5.2 Empirical evidence

Table 4 contains a heatmap that presents the selected variables and their relative importance (*gain contribution*) for each data set in our study. Each column of the heatmap represents the relative importance of the nine variables considered in our analysis. The darker shades in the table indicate higher variable importance. For example, Column 2 of the table shows the relative importance of the selected variables in the SJM 1 data set. These findings indicate that within the set of nine variables under consideration, COGS and EBITDA exhibit the most pronounced relative importance, whereas NI and RE demonstrate the least relative importance when it comes to identifying and detecting instances of EM.

|       | Accuracy |           | Precision |           | Sensitivity |           | F1 score |           |
|-------|----------|-----------|-----------|-----------|-------------|-----------|----------|-----------|
|       | AVG (%)  | STDEV (%) | AVG (%)   | STDEV (%) | AVG (%)     | STDEV (%) | AVG (%)  | STDEV (%) |
| SJM 1 | 74.27    | 16.47     | 71.61     | 18.15     | 69.19       | 15.59     | 69.35    | 17.28     |
| SJM 2 | 82.35    | 8.54      | 81.28     | 16.83     | 75.45       | 22.23     | 74.74    | 12.22     |
| SJM 3 | 81.53    | 10.84     | 87.86     | 6.80      | 86.43       | 15.06     | 86.41    | 9.02      |
| SJM 4 | 89.61    | 12.93     | 95.58     | 2.81      | 91.35       | 15.38     | 92.88    | 9.59      |
| MJM 1 | 75.71    | 16.94     | 69.83     | 19.22     | 66.66       | 16.03     | 67.12    | 18.28     |
| MJM 2 | 79.90    | 8.08      | 79.83     | 13.33     | 73.66       | 22.68     | 73.41    | 13.10     |
| MJM 3 | 82.77    | 12.64     | 90.68     | 4.77      | 87.41       | 16.61     | 88.21    | 9.94      |
| MJM 4 | 90.19    | 11.77     | 96.23     | 2.13      | 91.99       | 13.55     | 93.64    | 8.18      |

Source: Authors' own creation

**Table 3.**  
Classification  
performance of  
XGBoost models

It is worth noticing that results presented in Table 4 imply that the variables COGS, EBITDA, TD, TE and R&D have the most impact on predicting EM for both SJM and MJM data sets. Moreover, these five variables and the order of their relative importance appear to exhibit a high degree of consistency across the different data sets, which underscores their reliability and relevance in predicting EM. That is, including these variables into accrual models is likely to enhance the accuracy and effectiveness of their ability to detect EM.

COGS holds the greatest degree of relative importance among all data sets, which is not surprising given its critical role in accruals models. On average, COGS has a relative importance of 40.16%. COGS represents the direct expenses incurred in manufacturing and selling a company's products and is a fundamental component of its income statement. By incorporating COGS into accruals models, it becomes possible to identify instances of EM in firms that manipulate their COGS accounts to meet their earnings targets. Roychowdhury (2006) suggests that firms often resort to tactics like providing temporary price discounts, overproduction to decrease COGS and aggressively cutting discretionary expenses to boost profit margins and avoid losses. To detect such practices, Kothari *et al.* (2005) assert that integrating COGS into accruals models, such as the SJM and MJM, is crucial. Doing so can improve their effectiveness and enhance their ability to detect EM. Therefore, including COGS in these models provides deeper insights into a company's earnings quality, ultimately strengthening the accuracy of these models in identifying EM.

The findings indicate that managers manipulate COGS the most in four data sets, namely, SJM 1, SJM 4, MJM 1 and MJM 4, suggesting that they use both upward and downward manipulation techniques, simultaneously. This contrasts with findings which suggest that Chinese firms typically manipulate COGS in a single direction to manage earnings. Our study, however, suggests that some managers may use multiple techniques, at the same time, to manipulate COGS to achieve their desired earnings targets.

With a mean relative significance of 11.53%, the second most significant variable is EBITDA, which indicates a company's operational cash flow and its capacity to generate funds for meeting its debt responsibilities. However, EBITDA is often used as a proxy for cash flow, making it vulnerable to manipulation. Firms may artificially inflate or deflate revenues or expenses, by using aggressive revenue recognition or delaying expenses, to manipulate EBITDA, resulting in misleading information for investors and stakeholders.

Therefore, detecting EBITDA manipulation is crucial for evaluating a company's true operating performance and making informed investment decisions. Chen *et al.* (2021) found that Chinese firms engage in EM activities that significantly impact their reported EBITDA performance. EBITDA is particularly susceptible to manipulation due to its significance in

**Table 4.**  
Heat map of variable importance (*gain contribution*) in the developed XGBoost models across the different data sets

| Variable | SJM 1  | SJM 2  | SJM 3  | SJM 4  | MJM 1  | MJM 2  | MJM 3  | MJM 4  | Avg.   |
|----------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Year     | 0.0714 | 0.0915 | 0.0728 | 0.0631 | 0.0691 | 0.0857 | 0.0732 | 0.0625 | 0.0737 |
| TD       | 0.0831 | 0.0874 | 0.0860 | 0.0679 | 0.1120 | 0.0848 | 0.1296 | 0.0877 | 0.0923 |
| TE       | 0.0782 | 0.0983 | 0.0780 | 0.0426 | 0.0886 | 0.1297 | 0.0905 | 0.0551 | 0.0826 |
| RE       | 0.0501 | 0.0633 | 0.0513 | 0.0499 | 0.0487 | 0.0626 | 0.0507 | 0.0521 | 0.0536 |
| COGS     | 0.4080 | 0.3716 | 0.3850 | 0.5184 | 0.3890 | 0.3060 | 0.3582 | 0.4765 | 0.4016 |
| R&D      | 0.0932 | 0.0816 | 0.1017 | 0.0661 | 0.0728 | 0.0805 | 0.0741 | 0.0793 | 0.0812 |
| EBITDA   | 0.1147 | 0.1086 | 0.1228 | 0.0961 | 0.1169 | 0.1503 | 0.1157 | 0.0974 | 0.1153 |
| EBIT     | 0.0604 | 0.0579 | 0.0568 | 0.0663 | 0.0615 | 0.0575 | 0.0603 | 0.0618 | 0.0603 |
| NI       | 0.0410 | 0.0398 | 0.0457 | 0.0297 | 0.0413 | 0.0429 | 0.0477 | 0.0277 | 0.0395 |

**Source:** Authors' own creation

debt covenant calculations and its use as a measure of a company's operating performance by investors and analysts. Table 4 shows that managers engage in EM in SJM 3, SJM 4, MJM 3 and MJM 4 data sets, indicating the potential for EBITDA manipulation, especially in the downward direction.

Furthermore, our analysis indicates that total liabilities and debt (TD) can have a significant impact on the ability of accruals-based models in detecting EM. The findings displayed in Table 4 indicate that TD holds the third most significant level of relative importance, with an average relative significance of 9.23%. Prior research has indicated that managers can manipulate earnings using debt as accruals in various ways. First, through accrual accounting, managers can recognize revenue and expenses before they are actually paid or received, manipulating earnings in the short term (Bartov, 1993; Guidry *et al.*, 1999). Second, managers can adjust loan loss reserves, provisions set aside to cover potential losses on loans, through under-provisioning or over-provisioning (El Diri *et al.*, 2021). Third, managers can capitalize deferred financing costs, expenses associated with obtaining debt financing and amortize them over time to artificially reduce their impact on earnings in the short term (He and Zhou, 2021; Ettredge and Huang, 2019). Fourth, restructuring or repaying debt can lead to a reduction in interest expenses, thereby increasing the reported earnings (Moehrle, 2002; Ghouma, 2017; Mamatzakis *et al.*, 2023).

Results displayed in Table 4 also suggest that out of the nine financial variables under examination, Total Shareholders' Equity (TE) holds the fourth highest significance in recognizing instances of EM (8.26% average relative significance). TE constitutes a fundamental component of a company's financial posture, exerting considerable influence on the depiction and interpretation of earnings. One avenue of manipulating total equity entails strategically adjusting the timing of revenue and expense recognition. Through the deliberate postponement of expenses or the acceleration of revenue recognition, companies can artificially elevate current earnings, imparting a perception of inflated profitability (Altamuro *et al.*, 2005; Marquardt and Wiedman, 2004).

Moreover, companies can manipulate total equity by establishing or releasing reserves and provisions (Blouin *et al.*, 2010). As an illustration, a company might augment its bad debt reserve, temporarily dampening reported earnings for the current period. Subsequently, this reserve can be released in a subsequent period, bolstering earnings as required. Another innovative approach involves the engagement in share buybacks or the issuance of new shares. The repurchase of shares diminishes the count of outstanding shares, amplifying earnings per share (EPS) and potentially stimulating stock prices. Conversely, the issuance of new shares dilutes ownership, thereby affecting EPS and potentially mitigating the impact of unfavorable earnings news (Hribar *et al.*, 2006; Farrell *et al.*, 2014). Manipulating total equity can also occur indirectly through off-balance sheet transactions, whereby certain assets or liabilities are omitted from the balance sheet to portray a more favorable financial status (Pfeiffer, 1998). Such transactions possess the capacity to influence earnings and total equity in covert manners.

Finally, our findings indicate that R&D holds the fifth highest significance in identifying instances of EM (8.12% average relative significance). The expenses associated with R&D are a crucial factor that can significantly impact the effectiveness of accruals models. These models are commonly used to estimate the amount of earnings that a company is yet to receive in cash flows. R&D expenses are part of discretionary accruals, which can be manipulated to influence a firm's financial performance. As a critical component of discretionary accruals, R&D expenses can significantly affect the accuracy of accrual estimates and the overall effectiveness of accruals models. High levels of R&D expenses may indicate an aggressive approach to EM, resulting in inaccurate predictions about future



financial performance (Shust, 2015; Prasetyo *et al.*, 2022). Results obtained from our ML classifier analysis suggest that R&D has a significant impact on the accuracy of detecting EM across all data sets in a relatively consistent manner. Thus, it is imperative to factor in R&D expenses when designing accruals models to ensure accurate financial predictions.

## 6. Conclusion

EM poses a significant challenge in financial reporting, attracting attention from academia, regulatory bodies and investors alike. By manipulating financial outcomes to achieve earnings targets, EM can distort financial statements, undermining investor trust. This study leverages ML to uncover and assess new factors influencing EM, pinpointing variables such as COGS, EBITDA, Total Debt (TD), Total Shareholders' Equity (TE) and R&D expenses as crucial in predicting EM activities. Our findings reveal the dynamic nature of accruals-based models in EM detection, influenced by the evolving reliability of accrual estimates.

The implications of this research are far-reaching. For investors, it underscores the necessity of vigilance regarding EM tactics and the factors influencing such behaviors, enhancing investment decision-making. Regulators and policymakers are encouraged to fortify regulatory frameworks and surveillance mechanisms to mitigate EM, thus reinforcing market integrity and investor confidence. The study also demonstrates how auditors and forensic accountants can enhance audit quality and financial statement reliability by incorporating ML and specific financial indicators into their methodologies. Corporate managers are guided toward ethical financial reporting standards, highlighting the risks associated with EM practices.

Moreover, the research advocates for a holistic approach to combating EM, emphasizing stringent financial disclosures, enhanced audit practices incorporating analytics and robust corporate governance. It stresses the importance of regular financial reporting, strict penalties for noncompliance and minimized accounting latitude to deter subjective interpretations. The integration of ML for identifying financial anomalies, coupled with solid whistleblower protections and ethics education, is deemed essential. Additionally, international cooperation on standardizing and enforcing regulations is crucial for addressing the global challenge of EM, aiming to improve transparency, accountability and deter financial misconduct.

## Notes

1. A detailed discussion of the nine variables is provided in the following subsection, Section 4.2.
2. Details of this cross-validation process are elaborated in Section 5.

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